AMITY UNIVERSITY PUNJAB

B.Tech.(CSE)-4th Semester

Subject: Design and Analysis of Algorithms (CSE207)

Assignment 2

Q1. Derive the DP formula for LCS and find the longest common subsequence for strings X= ABCBDAB and Y=BDCABA.

Q2. Illustrate the algorithm using the dimensions:

A1(10 ×20). A2(20×50). A3(50×40). Show the m(i,j) and s(i,j) tables.

Q3. Describe the Backtracking paradigm. How does it prune the state-space tree compared to a simple exhaustive search?

Q4. Write the recursive algorithm for the N-Queens problem and explain the "Safe" function used to check row, column, and diagonal constraints.

Q5. Define the classes P, NP, NP-Hard, and NP-Complete. Draw a Venn diagram representing their relationship assuming P ≠ NP

Q6. What is Polynomial-Time Reduction? Explain its importance in proving that a new problem is NP-Complete.

Q7. Discuss Draw the portion of state space tree generated by FIFOBB for the following instance of 0/1 knapsack n= 5, M=12, (p1, ……p5) = (10,15,6,8,4) (w1,. ...w5)=(4,6,3,4,2)[14M]

Q8. What is non deterministic algorithm explain.